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PALAEOMAGNETISM OF THE DECCAN TRAP FLOWS OF JALNA, INDIA

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Palaeomagnetism of three Deccan Trap flows from Jalna (Maharashtra State, India) has been studied. These flows belong to the compound-lavas in the Deccan Traps and were sampled in an elevation range of 509 to 518 m m.s.l. The stable remanence of these flows, confirmed after magnetic cleaning tests, was found to be of reversed polarity (mean declination, 160°E; mean inclination, 46° down). The directional changes between these flows are systematic and can be attributed to secular variation of the palaeogeomagnetic field.

1. Introduction

Palaeomagnetic investigations have been made on the Deccan Trap lava formations of peninsular India by several workers [1–3]. The results of these studies, however, call for further intensive investigations on these lava sequences. Studies made on the Deccan lavas of Jalna (19°50′N, 75°56′E), about 160 km south-east of the renowned Ellora caves in the Maharashtra State, are described here. These basalt flows are dominantly vesicular and are devoid of inter-trappean formations, three of which were sampled and are described in table 1.

Table 1
Deccan Trap flows sampled in Jalna.

Flow	Elevation above the m.s.l. (m)	Rock type	No. of oriented samples collected
JC	518 –515	Vesicular	14
JB	515 –510.5	Amygdaloidal	14
JA	510.5–509	Vesicular	14

The top of flow JC and the bottom of flow JA could not be observed.

2. Measurements and results

The natural remanent magnetism (NRM) of the cylindrical specimens (2.5 cm diameter and same length cut from the samples) was measured using an astatic magnetometer. Progressive demagnetization of some of the specimens, in alternating fields up to 600 Oe amplitudes, proved the stability of the primary remanence in these lavas. At least one specimen from each sample was then directly demagnetised in 300–400 Oe in order to obtain magnetically clean directions. These directions before and after demagnetization, are shown on Schmidt projections in fig. 1 and the corresponding statistical data are summarised in table 2. The palaeolatitude (λ) and the virtual geomagnetic pole (VGP) position calculated from the cleaned remanence data are given in table 3.

It was found that the distribution of the NRM intensity (J_{nrm}) values of these samples could be best represented by a lognormal (fig. 2) rather than by a normal distribution model. These values showed no systematic variation from one flow to another and ranged between 8.5 and 34.2×10^{-4} emu/g.

3. Discussion

Examination of the flow mean directions after

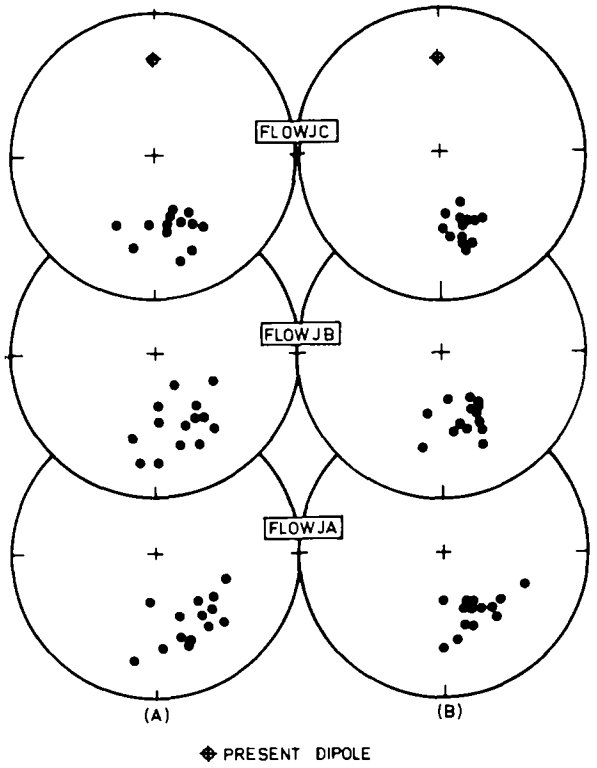


Fig. 1. Sample mean remanent magnetic directions of the Deccan Trap flows of Jalna. Schmidt projection (A) before, and (B) after magnetic cleaning in 300–400 Oe. All dips down.

magnetic cleaning clearly shows that the mean declinations change clockwise from the lower towards the upper flows (from 151° in the lower to 167° in the upper flow) while the corresponding inclinations also appear to register a regular decrease (from + 47° in the lower flow to + 44° in the upper). The significance of this systematic directional variation can be appreciated from the analyses of within- and between-flow directional dispersions in these lavas. In case the within-flow dispersions exceed the between-flow dispersions at some significance level (say P = 0.05), it would indeed be quite meaningless to ascribe any prominence to the directional changes between flows.

In order to test this relative significance of the directional dispersions, the method of analysis of dispersions [4] has been adopted as given in table 4.

The significance of the between-flow variation can then be judged by an F-test as follows:

$$\frac{\text{Mean square between-flows}}{\text{Mean square within-flows}} = F_2(B-1)\,,$$

$$2\Sigma(W_i-1) = F_{4,\,78}\,.$$

For the above data, the statistic $F_{4,\,78} = 2.586$ exceeds the variance ratio at P = 0.05. Hence the between-flow dispersion can be considered to be significant at this probability level. This suggests that the difference between the mean directions observed

Table 2
Statistical analysis of remanent magnetic data of Jalna Deccan Traps.

Flow	N	M_n						M_{af}				
		D	I	R	k	A_{95}	J_{nrm}	D	I	R	k	A_{95}
JC	14	174	+ 45	12.82	11.02	11.30	17.2	167	+ 44	13.64	36.1	6.2
JB	14	162	+ 43	13.19	16.05	9.36	18.1	162	+ 46	13.55	28.9	7.0
JA	14	159	+ 43	13.22	16.67	9.20	17.9	151	+ 47	13.68	40.6	5.9
Mean	42	165	+ 44	39.08	14.04	5.80	17.7	160	+ 46	40.72	32.0	3.8

M_n = data before magnetic cleaning
M_{af} = data after magnetic cleaning in 300–400 Oe
N = number of samples per flow
D = mean declination (E of true N)
I = mean inclination (+ ve. down)
R = resultant of N unit vectors
k = Fisher's precision parameter [$k = (N-1)/(N-R)$]
A_{95} = radius of 95 per cent circle of confidence
J_{nrm} = intensity of natural remanent magnetisation (lognormal mean value) in units of 10^{-4} emu/g.

Table 3
The VGP and related parameters corresponding to the cleaned RM directions of the Jalna Traps; $\lambda_0 = 19.9^\circ N$, $L_0 = 75.9^\circ E$.

Flow	λ_s	ψ	VGP (N pole)			
			$\lambda_p N$	$L_p W$	δ_p	δ_m
JC	26.1	−13.0	42.3	88.2	4.9	1.3
JB	27.4	−18.0	39.6	83.2	5.7	1.3
JA	29.2	−29.0	34.4	72.9	4.9	1.3

λ_0, L_0 = present latitude and longitude of the site
λ = ancient latitude of the site, i.e., $\lambda = \tan^{-1}(\frac{1}{2}\tan I)$
ψ = rotation of the site (clockwise positive)
λ_p, L_p = latitude and longitude of the Virtual Geomagnetic Pole (VGP)
δ_p, m = errors in estimation of VGP [11].

in the three flows is independent of the directional dispersions within the individual flows.

This systematic directional change can be attributed to either polar wandering and/or continental drift, or alternatively to secular variation of the geomagnetic field or to a compound effect of these. It should be noted here, however, that the basalt lavas in this neighbourhood are compound-flows [5, 6]. This implies that the individual flow-units were extruded in quick succession within a fairly short duration of time. This can be inferred from the lack of any irregularities such as traces of denudation, irregular contacts between successive flows, inter-trappean beds etc. and the thinness of the individual flows (i.e., flow-units, to use the original definition of Nichols [7]).

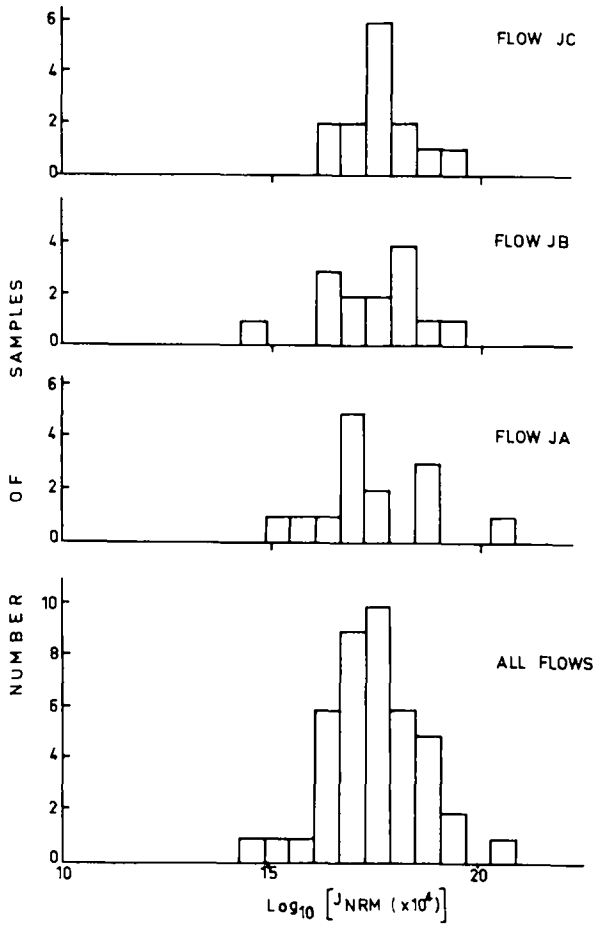


Fig. 2. Sample NRM intensities of the Deccan Trap flows of Jalna on lognormal distribution model.

Table 4
Analysis of directional dispersion in Deccan Traps of Jalna.
Number of flows (B) = 3; number of samples per flow (W_i) = 13; total number of samples (N) = 42; overall resultant vector of samples (R) = 40.72; resultant vector of samples flow by flow $\Sigma(R_i)$ = 40.87.

Source	Degrees of freedom	Sum of squares	Mean square
Between-flows	$2(B-1) = 4$	$\Sigma R_i - R = 0.15$	$\frac{\Sigma R_i - R}{2(B-1)} = 0.0375$
Within-flows	$2\Sigma(W_i - 1) = 78$	$\Sigma(W_i - R_i) = 1.13$	$\frac{\Sigma(W_i - R_i)}{2\Sigma(W_i - 1)} = 0.0145$
Total	82	1.28	

The possibility of this directional change having been caused by continental drift and/or polar wandering envisages either a rapid drift rate for the landmass or a long time span for the extrusion of these lavas. However, the investigations made so far on the Deccan lavas have not recorded such a drift rate. While the general palaeomagnetic results from Deccan Traps support the concept of the northerly drift of Indian landmass during this period, it is indeed difficult to attribute the directional changes in Jalna flows to the same cause in view of the time span of their extrusion. Similarly the concept of polar wandering, either alone or in conjunction with drift, would be a remote possibility. It is very probable, therefore, that the changes in the directions observed in the Jalna flows were caused by the secular variation of the ambient geomagnetic field.

Following the arguments put forth by Clegg et al. [8], it can be inferred that these Jalna flows were magnetized when the Indian landmass was in the southern hemisphere. A comparison of the present results with those of the other Deccan Trap flows [1, 3, 9] reveals that the Jalna flows are directionally closer to those of Linga ($D = 164°, I = +48°$ [8]). This is quite suggestive of the contemporaneity of the lavas in these two areas [10]. However, unlike the Jalna flows, the Linga flows show no regular directional changes in the successive flows. It may be noted here that the former are compound-flows while the latter are simple-flows. Further, while the three Jalna flows studied comprise a vertical thickness of only 10 m, the four flows studied in Linga account for about 100 m of vertical thickness. Thus while the flows in Jalna and Linga areas can be broadly taken to be contemporaneous, the former possibly occupied relatively very small geological time span. As a result, the secular variation of the ambient geomagnetic field could have been systematically recorded by the magnetization of Jalna flows while in the Linga flows, which extruded over a longer period of time, the direction of magnetisation must have averaged out.

Acknowledgements

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